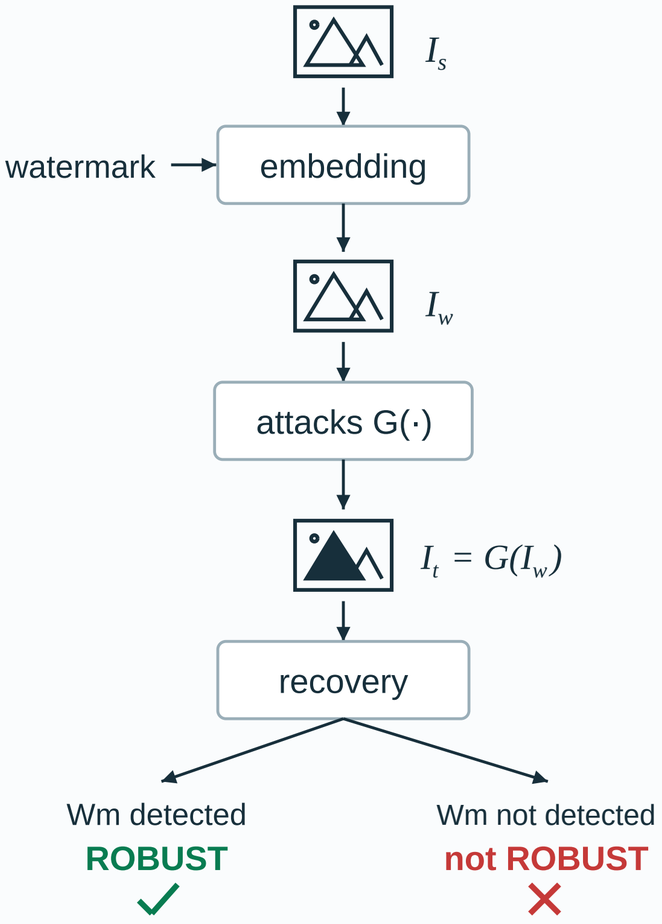
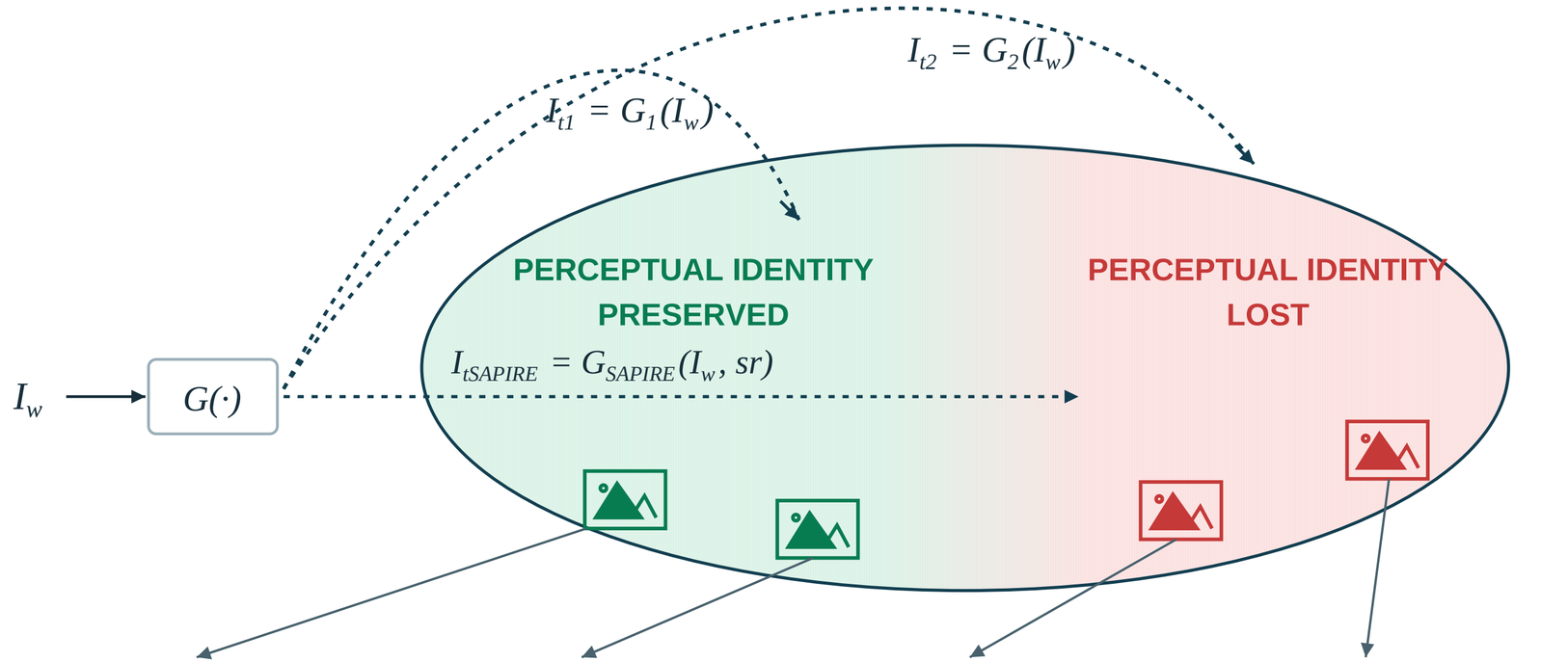
Rethinking Robustness in Digital Watermarking: A Perceptual Identity-Based Framework

Graphical abstract by Angela D'Angelo

Robustness evaluation in digital watermarking



Rethinking robustness through the perceptual identity transition region and Semantic-Aware Perceptual Identity Region Editing (SAPIRE)



Current evaluations do not consider the semantic aspect of the attacks. Robustness is evaluated based on watermark detectability.

Conceptual illustration of the perceptual identity framework. The Perceptual Identity Boundary (PIB) captures the transition region between preserved and lost perceptual identity. Semantic-Aware Perceptual Identity Region Editing (SAPIRE) enables controlled exploration within this region.